

MUHAMMAD GHIFARI PERMATA ALWY

QUALITY ASSURANCE ENGINEER | SYSTEM ANALYST | TECHNICAL WRITER
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SUMMARY

A detail-oriented and adaptable Quality Assurance Engineer with a strong foundation in software engineering, system analysis, and enterprise technical documentation. Experienced in designing comprehensive test plans, executing manual and automated testing (Vitest, PHPUnit), setting up CI/CD quality gates (GitHub Actions, SonarCloud), and validating microservices, API gateways (WSO2 APIM/IS), and event-driven systems (Apache Kafka). Skilled in translating complex business requirements into Requirement Traceability Matrices (RTM), High/Low-Level Designs (HLD/LLD), and testable acceptance criteria to ensure 100% requirement coverage, defect prevention, and reliable, high-quality digital products.

PROFESSIONAL EXPERIENCE

Internship, IT Support Specialist – Project and Service Delivery 2 Division, PT Infrastruktur Telekomunikasi Indonesia **July 2025 - Sep 2025**

- Developed and performed functional quality assurance on DASH-DISM, a web-based Dashboard Analyst prototype for monitoring the dismantling of microwave antennas on the 1879 HOP project, using React.js/TypeScript for the frontend and Express.js for the backend.
- Designed, implemented, and verified role-based authentication and access control (RBAC) test scenarios across three user roles (Project Manager, Field Coordinator, and Data Entry User) to ensure strict authorization boundaries.
- Built and validated a collaborative data entry portal for daily plan (morning) and daily actual (evening) reporting, eliminating manual data entry errors from Notepad and WhatsApp group recaps.
- Verified data accuracy and calculation logic across real-time S-Curve visualizations, weekly dismantling quantity charts, and activity progress breakdowns by weighted KPI.
- Tested automated reporting workflows to generate daily plan and actual reports, validating data management features including Excel import validation, data export, and audit logs.
- Provided hardware, software, and network troubleshooting and root-cause analysis (RCA) support for internal staff, ensuring minimal downtime and smooth day-to-day operations.
- Maintained and audited the Asset Tagging and Tracking application, verifying data accuracy and operational reliability across the division's asset records.
- Prepared technical test reports and implementation documentation to support defect tracking, project delivery, and knowledge transfer within the team.

System Analyst – Yunas Studio **Sep 2025 - July 2026**

- Conducted requirements engineering and testability analysis through stakeholder interviews and field observation with Yunas Studio's admin, resulting in 40 Functional Requirements and 14 Non-Functional Requirements with testable acceptance criteria across three user roles (Customer, Admin, Superadmin).
- Modeled the existing (as-is) business process using the Functional and Object-Oriented Methodology (FOOM), producing a business Data Flow Diagram (DFD) and an Entity Relationship Diagram (ERD) covering 10 entities including Booking, Payment, Schedule, and Package to verify data integrity.
- Designed and validated system behavior using Object-Oriented Analysis and Design (OOAD), producing 23 Use Cases, 23 Activity Diagrams, 23 Sequence Diagrams, and a Class Diagram with 9 core classes and 5 relationship types to serve as functional and scenario-based test specifications.
- Designed and verified a layered system architecture (Frontend, Backend, External Services, Database) for a Progressive Web App (PWA), including integration workflows with a Payment Gateway (Midtrans/Xendit) to automate payment verification and eliminate manual confirmation errors via WhatsApp.
- Authored a Software Requirements Specification (SRS) and Software Design Document (SDD), mapping all 54 requirements to design and verification artifacts through a Requirement Traceability Matrix (RTM), achieving 100% requirement and test coverage.
- Facilitated User Acceptance Testing (UAT) and stakeholder validation sessions with Yunas Studio's admin, achieving 100% approval across 10 core business process aspects and confirming the system met actual operational needs.
- Bridged communication between business stakeholders and the development team, and collaborated with UI/UX designers to validate proposed solutions against real business workflows.

Full Stack Developer (Freelance), CV. Permata Alam Perkasa **Aug 2026 – Okt 2026**

- Developing, customizing, and performing end-to-end functional quality assurance on an ERP system using the Odoo Framework to support business operations of a retreaded and truck tire distribution company.
- Configuring and validating the Inventory module to verify accurate stock levels, multi-warehouse locations, and product movement tracking.
- Setting up and testing the Sales and Purchase modules to verify product distribution workflows, incoming and outgoing goods, and vendor/customer transaction integrity.

- Implementing and reconciling the Accounting module to ensure financial recording and reporting accuracy aligned with business processes.
- Configuring and verifying the Employees/Attendance module to manage employee master data and validate daily attendance tracking rules.
- Managing and auditing the Product module, validating product data, categories, and pricing structures across the ERP system.

Technical Writer – PT. Bentang Inspira Teknologi (Project SPLP) **Sep 2026 - Present**

- Authored comprehensive High-Level Design (HLD) and Low-Level Design (LLD) technical documentation for the National Government Service Integration System (Sistem Penghubung Layanan Pemerintah / SPLP), establishing clear architectural baselines, system specifications, and acceptance criteria for engineering and QA teams.
- Documented end-to-end SPLP Portal Management specifications across Frontend and Backend microservices, detailing UI/UX workflows, role-based access control (RBAC), RESTful API contracts, data models, and state transitions to support functional and integration testing.
- Produced detailed integration, routing, and policy documentation for WSO2 API Manager (WSO2 APIM) and WSO2 Identity Server (WSO2 IS), defining API gateway throttling/rate-limiting rules, OAuth2/OIDC authentication flows, and centralized identity governance.
- Authored technical architecture, message routing flows, and capacity planning documentation for the event-driven platform powered by Apache Kafka (KRaft mode) and Go-based Message Exchange Service (MES), specifying topic partitioning, schema validation, and fault-tolerance benchmarks for load and reliability testing.
- Documented the full-stack monitoring and observability pipeline utilizing Grafana, Grafana Alloy, Prometheus, Jaeger, and ClickHouse, defining metrics collection, distributed tracing, and centralized audit logging standards to monitor system health and SLA compliance.
- Developed technical security and network infrastructure documentation covering Site-to-Site/Client VPN access controls, mTLS encryption, Wazuh SIEM security monitoring, and Intrusion Detection Systems (IDS) to support security auditing and vulnerability verification.

PROJECTS

Full Stack Developer – StudyIT & StudyIT Mobile **Sep 2023 - June 2024**

- Developed, maintained, and tested cross-platform web and mobile application features using React.js and Flutter.
- Built and verified responsive user interfaces across both web and mobile platforms for cross-device compatibility.
- Implemented and validated MVC (Model-View-Controller) architecture on the web backend using Laravel, separating Eloquent models (Course, Instructor, Enrollment, Discussion, Account) from role-based Blade views (Admin and Regular User) and dedicated controller classes.
- Built and tested a RESTful API service with Express.js following an OOP-based MVC pattern, verifying endpoints and payloads across dedicated resource controllers (Course, Account, Instructor, Enrollment, Discussion, Task, Reply, Item, Module).
- Implemented and tested authentication and security features, including JWT-based auth, two-factor authentication (OTP), and secure file upload validation (Multer).
- Collaborated on learning platform architecture, functional verification, and content delivery reliability.

Restaurant Order Management System – WinForms Desktop App **June 2024**

Back-End & Front-End Developer | Academic Project (Konstruksi Perangkat Lunak)

- Built and tested a desktop-based restaurant order management system using C# and Windows Forms (.NET Framework), following the MVC (Model-View-Controller) architecture to separate business logic, UI, and data layers for testability.
- Developed and verified the Login and Homepage modules, implementing strict input validation, boundary checks, and defensive coding practices (null/empty checks, structured exception handling) to prevent invalid data submission and application crashes.
- Implemented and validated JSON-based data persistence using System.Text.Json for reliable reading and writing of user account data.
- Modeled and verified order processing logic using a table-driven state machine (Pending → Processing → Completed/Cancelled) with generic classes and Design by Contract (DbC) assertions to enforce valid state transitions and prevent logical defects.

ShelfLink – Digital Library Management System | Full-Stack Developer **Sep 2025 - June 2026**

Academic Startup Project

- Developed and validated a web-based library management system to digitalize book cataloging, borrowing, returning, and review processes.
- Set up an automated CI/CD pipeline using GitHub Actions to run automated PHPUnit test suites on every commit, integrated with SonarCloud for continuous static code quality analysis, defect detection, and test coverage tracking.
- Participated in MVP development and led functional testing, regression testing, and usability testing, driving iterative product quality improvements based on user feedback.

- Designed and implemented frontend and backend features using Laravel and MySQL following an MVC architecture with dedicated model and controller classes (Book, Borrowing, BookReview, User, Catalog, Return, Statistics).
- Integrated and tested an AI-based image analysis service to automatically detect book damage from return photos, validating confidence score thresholds and suggested fine accuracy.
- Built and verified an automated fine calculation system combining overdue penalties and AI-suggested damage fines, with automatic overdue status detection and boundary-case validation.
- Built and tested CRUD modules for book management, user management, borrowing, and return transactions, along with role-based access control (RBAC) for administrators and library users.
- Implemented and verified automated email notifications (borrow confirmation, overdue reminders, return receipts) using Laravel Mail, and optimized database queries and RESTful workflows for system performance.

Full Stack Developer | Aplikasi Pendaftaran Beasiswa Pelatihan (Recruitment PT. Bentang Inspira Teknologi)
September 2026

- Designed, built, and verified a 6-service microservices architecture (API Gateway, RBAC, Master Data, Transaction, Document, Frontend) using Node.js 20, TypeScript, Express, Prisma, and React 18, orchestrated via Docker Compose.
- Wrote automated unit and integration tests using Vitest covering authentication, refresh-token rotation, and file-type validation, and produced full QA and technical documentation (ERD, OpenAPI spec, security audit, architecture decision records).
- Implemented and tested asymmetric JWT authentication (RS256) with private/public key separation between the auth service and gateway, argon2id password hashing, and refresh-token rotation with token reuse detection.
- Enforced and validated a private-network security model where all backend services and databases are isolated within a Docker internal network, accessible only through a single API Gateway with rate limiting, CORS whitelisting, and Helmet security headers.
- Built and security-tested a document upload pipeline with magic-byte file-type validation, ClamAV antivirus scanning, UUID-based storage, and SHA-256 checksums to prevent file-spoofing and unauthorized access.
- Designed and verified a database schema across MySQL (RBAC, Master, Transaction) and PostgreSQL (Document, using JSONB metadata), covering 14 tables with a state-machine-driven registration workflow (draft → submitted → verified → interviewed → passed).

EDUCATION

Bachelor of Software Engineering (S1 Rekayasa Perangkat Lunak) **2022 - 2026**
School of Computing, Telkom University
GPA: 3.38/4.00

SKILLS

Software Quality Assurance (SQA), Manual & Automated Testing, Unit & Integration Testing (Vitest, PHPUnit), API & Gateway Testing (RESTful API, OpenAPI/Swagger, Postman, WSO2 APIM, WSO2 IS), CI/CD & Code Quality (GitHub Actions, SonarCloud, Docker), Technical & Test Documentation (Test Plan, Test Cases, RTM, UAT, SRS, HLD, LLD), Event-Driven & Distributed Systems (Apache Kafka KRaft, Go MES, Microservices), Monitoring, Observability & Security (Grafana, Grafana Alloy, Prometheus, Jaeger, ClickHouse, Wazuh SIEM, IDS, ClamAV, VPN/mTLS), System & Requirement Analysis (OOAD, UML, BPMN, DFD, ERD), Full-Stack Development (React.js, TypeScript, Node.js, Express.js, Laravel, Flutter, Odoo ERP, MySQL, PostgreSQL)

ADDITIONAL INFORMATION

- Languages: Bahasa, English
- Certifications: Hak Kekayaan Intelektual (HKI) – Sistem Booking pada Foto Studio berbasis PWA (Progressive Web App)
- IBM Full-Stack Software Developer (Coursera), 2025
- Meta Full-Stack Software Developer (Coursera), 2025
- IBM Backend with Node.Js and Express Developer (Coursera), 2026

ORGANIZATIONS

- Professional Development Staff - BEM KEMA Telkom University **2022 - 2023**
- Staff of External Division - HMRPL (Himpunan Mahasiswa RPL) **2023 - 2024**
- Chairman - PERMALA (Perhimpunan Mahasiswa Lampung) Telkom University **2024 - 2025**

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